

Pre-transaction Check-in

Show the screens in this exact order. Each section gives the frontend input and backend scoring.

1. Store category

Question: "Where are you planning to shop?"

Frontend field: store_category **Control:** single-choice list

Answer sent by frontend	Marketing score M
clothing	1.00
mall	1.00
online	0.75
groceries	0.75
other	0.25
electronics	0.00
restaurant	0.00

Scoring: The backend looks up M from the table. The frontend sends only store_category; it does not calculate M.

2. Valence

Question: "How pleasant or unpleasant do you feel right now?"

Frontend field: valence_z **Control:** five choices

Answer	Label	Valence score ZV
1	Very negative / unpleasant	-2
2	Negative	-1
3	Neutral	0
4	Positive	+1
5	Very positive / pleasant	+2

Scoring: Map the selected choice directly using the table and send valence_z. The backend accepts only -2, -1, 0, +1, or +2 and uses the value unchanged. Do not subtract 3 again. Valence means pleasant versus unpleasant; it does not mean calm versus stressed.

3. Heart rate - automatic

No question is shown. The wearable sends current_hr_bpm. The backend already has baseline_hr_mean and baseline_hr_sd for the user.

```
z_hr_raw = (current_hr_bpm - baseline_hr_mean) / baseline_hr_sd
z_hr = clamp(z_hr_raw, -2.0, +2.0)
```

Scoring: Higher than the user's baseline gives a positive value. Lower than baseline gives a negative value. Reject the calculation if baseline_hr_sd is zero or missing.

Shared scale for all four sliders

Value	Meaning
-2.0	Extremely below baseline
-1.5	Very low
-1.0	Low
-0.5	Slightly low
0.0	Baseline / medium
+0.5	Slightly high
+1.0	High
+1.5	Very high
+2.0	Extremely above baseline

Important: These nine values are meaning anchors. The slider accepts every 0.1 value from -2.0 through +2.0: 41 possible values in total.

4. Heartbeat steadiness slider

Question: "How steady and calm does your heartbeat feel compared with usual?" **Frontend field:** perceived_hrv. **Control:** slider. **Direction:** -2.0 = much less steady / more irregular; 0.0 = normal; +2.0 = much steadier / calmer. **Scoring:** 41 values from -2.0 to +2.0 in steps of 0.1. Transmit this value separately.

5. Sweating slider

Question: "How sweaty or clammy do you feel compared with usual?" **Frontend field:** perceived_eda. **Control:** slider. **Direction:** -2.0 = much drier; 0.0 = normal; +2.0 = much sweatier / clammier. **Scoring:** 41 values from -2.0 to +2.0 in steps of 0.1. Transmit this value separately.

6. Breathing slider

Question: "How fast or shallow is your breathing compared with usual?" **Frontend field:** perceived_respiration. **Control:** slider. **Direction:** -2.0 = much slower / deeper; 0.0 = normal; +2.0 = much faster / shallower. **Scoring:** 41 values from -2.0 to +2.0 in steps of 0.1. Transmit this value separately.

7. Skin warmth slider

Question: "How warm or flushed does your skin feel compared with usual?" **Frontend field:** perceived_skin_temperature. **Control:** slider. **Direction:** -2.0 = much colder; 0.0 = normal; +2.0 = much warmer / more flushed. **Scoring:** 41 values from -2.0 to +2.0 in steps of 0.1. Transmit this value separately.

8. Values transmitted separately

Do not calculate one combined arousal score. Send normalized heart rate and each slider value as separate fields.

Value	Where it comes from
M	Store-category lookup
valence_z	Mapped valence choice; already scored from -2 to +2
z_hr	Current heart rate normalized against the user's baseline
perceived_hrv	Heartbeat-steadiness slider
perceived_eda	Sweating slider
perceived_respiration	Breathing slider
perceived_skin_temperature	Skin-warmth slider